

Operating System

Lesson 19 — Functions, Kernel and Management

JKSSB Computer Awareness

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Defining the Operating System

- ▶ An **operating system (OS)** is the main **system software** that controls the whole computer.
- ▶ It sits between the **user** and the **hardware**.
- ▶ It makes the machine usable: without an OS, the hardware alone cannot run application programs.
- ▶ It starts the computer, runs programs, and manages every resource.

The one idea

The OS is the layer that provides a **user-friendly interface** so the user can work without knowing the hardware details.

The Two Jobs of an Operating System

- ▶ **Interface**: it lets the user give instructions and receive results in an easy way.
- ▶ **Resource manager**: it shares the CPU, memory, storage, and devices among all running programs.
- ▶ Every other program depends on the OS to talk to the hardware.

Interface and manager

The OS is at once the friendly face of the computer and the manager of all its resources.

Where the Operating System Lives

- ▶ The OS is stored permanently in **secondary storage** (such as an SSD or HDD).
- ▶ At startup it is loaded into **primary storage** (main memory).
- ▶ While the computer is running, the OS is **resident in primary storage**.
- ▶ It stays in main memory throughout the session so it can control the machine instantly.

Trap alert

A running OS is in **primary storage** (main memory), not in secondary storage and not on the CPU chip.

Core Functions at a Glance

Function	What it controls
Process management	Creating, scheduling, and finishing running programs
Memory management	Allocating main memory to programs
File management	Creating, reading, writing, and deleting files
Device management	Driving input/output devices via drivers
Security	Logins, accounts, permissions, updates
User interface	GUI or CLI for user interaction

The Kernel

- ▶ The **kernel** is the **core** of the operating system.
- ▶ It is the part that interacts **directly with the hardware**.
- ▶ It always stays in main memory while the computer runs.
- ▶ It provides the most important services: process scheduling, memory handling, and device control.

Kernel

The central, privileged part of the OS that manages hardware and provides the services other software depends on.

User Interfaces: GUI vs CLI

Point	GUI	CLI
Full form	Graphical User Interface	Command Line Interface
Input	Clicks, icons, windows	Typed text commands
Ease	Easier for beginners	Needs commands to be known
Example	Windows desktop, icons	Command prompt, terminal

GUI = Graphical User Interface; CLI = Command Line Interface.

Booting the Computer

- ▶ **Booting** is the process of starting the computer and loading the operating system into main memory.
- ▶ **Cold boot**: starting from the power-off state.
- ▶ **Warm boot**: restarting a computer that is already on.
- ▶ The firmware **BIOS/UEFI** first checks the hardware, then finds and loads the OS from secondary storage.

Who starts the boot

BIOS/UEFI initialises the hardware and hands control to the operating system.

Process Management

- ▶ A **process** is a program that is currently executing.
- ▶ The OS creates processes, gives each a turn on the CPU, and removes them when they finish.
- ▶ **CPU scheduling** decides which process runs next and for how long.
- ▶ It also allows processes to communicate and protects them from each other.

Program vs process

A program on disk is passive; once it is loaded and running it becomes an active process.

Memory Management

- ▶ The OS decides **how much main memory** each program gets.
- ▶ It keeps one program's memory **separate** from another's.
- ▶ It frees memory when a program ends, so it can be reused.
- ▶ It may use **virtual memory** to give programs more space than the physical RAM provides.

Why it matters

Without memory management, two programs could overwrite each other's data.

File Management

- ▶ The OS organises data into **files** and **folders**.
- ▶ It handles creating, reading, writing, renaming, and deleting files.
- ▶ It keeps a **directory** that records where each file is stored.
- ▶ It controls **permissions**: who may open, change, or delete a file.

A file system

The way the OS names, stores, and finds files on a storage device.

Device Management

- ▶ The OS controls all **input/output devices**: keyboard, mouse, printer, disk, and so on.
- ▶ Each device talks to the OS through a **device driver**.
- ▶ A device driver is **software**, not hardware.
- ▶ **Spooling** lets the OS queue print jobs and feed the printer at its own speed.

Trap alert

A device driver is **software**; the operating system itself is **system software**.

Multitasking and Time-Sharing

- ▶ **Multitasking** means the OS runs **more than one program at a time**.
- ▶ The CPU switches between programs so quickly that they appear to run together.
- ▶ **Time-sharing** lets **many users** share one computer at the same time, each getting a slice of CPU time.
- ▶ Both rely on the OS **scheduler**.

Trap alert

Multitasking = running more than one program at a time; time-sharing = sharing one machine among several users.

Security, User Accounts and Updates

- ▶ The OS provides **user accounts** with a username and password.
- ▶ **Authentication** (login) verifies who is using the computer.
- ▶ **Permissions** decide which files and settings each account may use.
- ▶ Regular system and software **updates** fix security holes and add improvements.

Common Operating Systems

- ▶ **Microsoft Windows**: widely used desktop and laptop OS.
- ▶ **Linux**: an open-source OS family; Android is built on it.
- ▶ **macOS**: Apple's desktop and laptop OS.
- ▶ **Android** and **iOS**: mobile operating systems.

An application such as MS Word is not an operating system; it runs on top of one.

Trap Alert: OS Facts to Keep Straight

- ▶ The OS is **system software**, not application software. (TRAP-10)
- ▶ While running, the OS is resident in **primary storage** (main memory). (PYQ-22)
- ▶ The OS gives a **user-friendly interface** between user and hardware. (PYQ-21)
- ▶ **Multitasking** = running more than one program at a time. (PYQ-23)
- ▶ **BIOS/UEFI** initialises hardware and loads the OS; it is not the kernel or a device driver. (TRAP-24)
- ▶ A device driver is **software**, not hardware. (TRAP-10)

Lesson 19: Recall

- ▶ The OS is system software that provides a user-friendly interface and manages all resources.
- ▶ It is stored on secondary storage but is resident in **primary storage** while running.
- ▶ Core functions: process, memory, file, and device management, plus security and the user interface.
- ▶ The **kernel** is the core part that talks directly to hardware.
- ▶ **GUI** = Graphical User Interface; **CLI** = Command Line Interface.
- ▶ **Booting** loads the OS; BIOS/UEFI starts the process.
- ▶ **Multitasking** runs more than one program at a time; **time-sharing** shares one machine among many users.
- ▶ User accounts, authentication, permissions, and updates provide security.